4-G-Array Sum max problem

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer:(penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b)
{
   return (*(int *)a - *(int *)b);
}

int main()
{
   int n;
   scanf("%d", &n);

   int arr[n];

   for (int i = 0; i < n; i++)
   {
      scanf("%d", &arr[i]);
   }

   // Sort in ascending order
   qsort(arr, n, sizeof(int), compare);

   long long sum = 0;
```

```
    for (int i = 0; i < n; i++)
    {
        sum += (long long)arr[i] * i;
    }

    printf("%lld", sum);

    return 0;
}
```

Feedback

	Input	Expected	Got	
	5 2 5 3 4 0	40	40	
	10 2 2 2 4 4 3 3 5	191	191	

	Input	Expected	Got	
	5 5			
	2 45 3	45	45	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.